

Introduction to Fault-Tolerant Quantum Circuits

AlgoLab Summer School — Lecture 1

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Prelude: Why Build a Quantum Computer?

A Quantum Industry Snapshot

- **Over \$10B** in funding from 2012-2024.
 - **\$2B** recently announced by the US Government (CHIPS and Science Act).
- **24+ companies** with commercially available Quantum Processing Units.
- **16000+** full time employees at “pure-play” quantum companies.
- **1B+** in industry revenue in 2025.

Sources: MIT Quantum Index Report 2025 (qir.mit.edu);

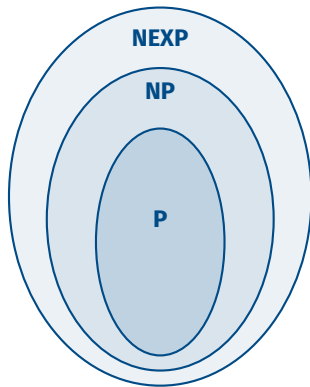
QED-C State of the Global Quantum Industry 2026 (quantumconsortium.org)



Photo: IBM Research / CC BY-ND 2.0

Where It All Began: The Extended Church-Turing Thesis

- The **Church-Turing Thesis**: any function computable by a physical process can be computed by a Turing machine.
- The **Extended Church-Turing Thesis**: any function *efficiently* computed by a physical process can be *efficiently* computed by a Turing machine.
- **Reasoning**: Any physical process can be efficiently simulated by a Turing machine. So if a physical process efficiently computes a function, the Turing Machine can simulate that process, then read the result of the computation.

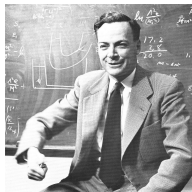


Quantum Enters the Picture

- From the Extended Church Turing reasoning: “Any *physical process* can be **efficiently simulated** by a Turing machine.”
- What equations could be used to describe our physical process?
 - Classical mechanics ✓ — efficiently simulable
 - Electrodynamics ✓ — efficiently simulable
 - Special relativity ✓ — efficiently simulable
 - Quantum mechanics ✗ — does *not* appear efficiently simulable

“Nature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical.”

— Richard Feynman, 1981



The Quantum Potential

- A device which processes information using the laws of quantum mechanics *may not* be efficiently simulable by a Turing machine.
- Such a device *may* be able to efficiently solve problems which cannot be efficiently solved by any other classical physical process.

Part 1: Physical Qubits

One (Physical) Qubit States

- A (physical) **qubit** models the behavior of a **two-level** quantum system.



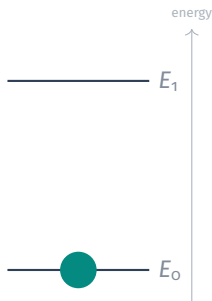
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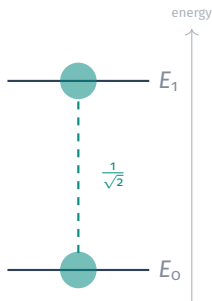
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 - $|\psi\rangle = |1\rangle$: the system is in the other (excited) state.
 - $|\psi\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$: The system is in a **superposition** of the ground and excited states.



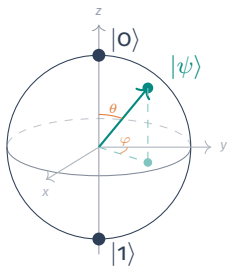
Math Recap

- A state of a qubit is a norm-one vector in some **Hilbert space**.
 - **Hilbert Space**: A vector space with inner product, complete in the induced norm.
 - For finite dimensions the unique choice (up to isomorphism) is $\mathbb{C}^n$ with the standard inner product.

- One qubit states are vectors in a two-dimensional Hilbert Space.

$$\text{i.e. } |\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

- A useful geometric representation of one qubit states is given by the **Bloch sphere**.



$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

One (Physical) Qubit Gates

- Physically, single-qubit gates evolve the state $|\psi\rangle$ according to the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle \quad \implies \quad |\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle$$

- Without loss of generality, this evolution looks like a **rotation** about some axis on the Bloch sphere.
- When engineering a physical qubit, we want to control the Hamiltonian to effect rotation around different axes.

Question: Assume we have enough control over our Hamiltonian to affect rotation about two different axes. Say (for mathematical convenience) we can rotate around X and Z:

$$R_X(\theta) = e^{-i\theta X/2}, \quad R_Z(\theta) = e^{-i\theta Z/2}$$

Then can we implement any single qubit unitary? (And how?)

Compiling a Single Qubit Unitary

Q: Can we implement any single unitary with R_X and R_Z gates? **A:** Yes!

Proof: First recall A unitary U is fully specified by where it sends basis vectors:

$$|0\rangle \mapsto |\psi\rangle, \quad |1\rangle \mapsto |\psi^\perp\rangle$$

Since the Bloch sphere represents states up to global phase, we can write:

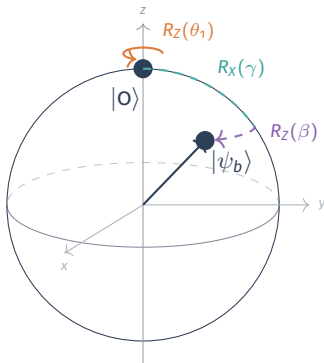
$$|0\rangle \mapsto e^{i\theta_1} |\psi_b\rangle, \quad |1\rangle \mapsto e^{-i\theta_1} |\psi_b^\perp\rangle$$

where $|\psi_b\rangle$ is now fully specified by a point on the Bloch sphere. We can then decompose U as:

$$U = R_Z(\gamma) \cdot R_X(\beta) \cdot R_Z(\theta_1)$$

where $R_Z(\theta_1)$ contributes the phase, and $R_X(\beta) \cdot R_Z(\gamma)$ rotates to the correct point on the Bloch sphere.

Compiling a Single Qubit Unitary



$$U = R_z(\beta) \cdot R_x(\gamma) \cdot R_z(\theta_1)$$

Note: This also shows that every single-qubit unitary can be represented by a rotation of the Bloch Sphere.

Many (Physical) Qubit States

- An n -qubit state is described by a vector in $\mathbb{C}^{2^n}$.
- If the qubits are independent, the state is a **product state**:

$$|\psi\rangle = |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$$

- Single-qubit unitaries send product states to product states:

$$U_1 \otimes \cdots \otimes U_n (|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle) = (U_1 |\psi_1\rangle) \otimes \cdots \otimes (U_n |\psi_n\rangle)$$

- To **entangle** qubits we need access to **multi-qubit unitaries**.

Bonus Question(s):

1. (Easy) How many real numbers does it take to describe an n -qubit state. An n -qubit product state?
2. (Harder) How many bits to describe an n -qubit state and n qubit product state to within error ε ?

From your answers to above, explain why we say product states can be described classically, while general states cannot.

Measuring Many Qubit States

- Recall:
 - A **projective valued measure** is a set of projectors $\{P_1, \dots, P_k\}$ summing to the identity.
 - Applying the corresponding measurement to state $|\psi\rangle$ returns outcome i with probability $\|P_i |\psi\rangle\|^2$, after which the system is in state $P_i |\psi\rangle / \|P_i |\psi\rangle\|$.
- One standard measurement made on n physical qubits is a **readout**: measuring $\{|x\rangle\langle x|\}_{x \in \{0,1\}^n}$
- But we can also perform many others! These need not be destructive! For example, we could have:

$$P_{\text{even}} = |00\rangle\langle 00| + |11\rangle\langle 11|, \quad P_{\text{odd}} = |01\rangle\langle 01| + |10\rangle\langle 10|$$

And applying the PVM $\{P_{\text{even}}, P_{\text{odd}}\}$ to the state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ returns outcome P_{even} with probability 1 and post-measurement state:

$$P_{\text{even}} |\Phi^+\rangle = |\Phi^+\rangle.$$

Many-Qubit Unitaries

- Say we can engineer a two-qubit interaction $H \propto ZZ$. Evolving for the correct time gives the **CZ gate**:

$$\begin{aligned} CZ |00\rangle &= |00\rangle, & CZ |01\rangle &= |01\rangle, \\ CZ |10\rangle &= |10\rangle, & CZ |11\rangle &= -|11\rangle \end{aligned}$$

- Equivalently, $CZ = \text{diag}(1, 1, 1, -1)$, or:

$$CZ = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z$$

Questions: Assume we only have access to the CZ gate and arbitrary single-qubit unitaries. Using only a sequence of these gates, can we implement:

1. (Hard) Any unitary of the form CU for arbitrary U ?
2. (Very Hard) Any two-qubit unitary?
3. (Very Hard) Any multi-qubit unitary?

Many-Qubit Unitaries: Proof 1

Q: Can we implement any CU gate using CZ and single-qubit gates?

A: Yes!

Proof Sketch:

- Find matrices A, B, C such that $ABC = I$ and $AZBZC = U$.
 - How? Let $U = R_Z(\beta)R_Y(\gamma)R_Z(\delta)$ be an Euler angle decomposition of U and set:

$$A = R_Z(\beta)R_Y(\gamma/2), \quad B = R_Y(-\gamma/2)R_Z(-\frac{\delta+\beta}{2}), \quad C = R_Z(\frac{\delta-\beta}{2})$$

(Exercise: Check this is correct)

- Then set $CU = (IA)(CZ)(IB)(CZ)(IC)$
- As a circuit:

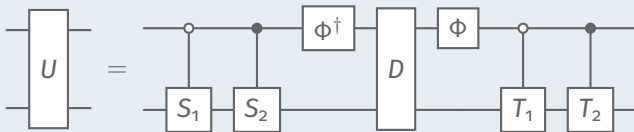


Many-Qubit Unitaries: Proof 2

Q: Can we implement any two-qubit unitary using CZ and single-qubit gates? **A:** Yes!

Proof Sketch:

- By the *sin-cosine decomposition* any two qubit unitary U can be written as:



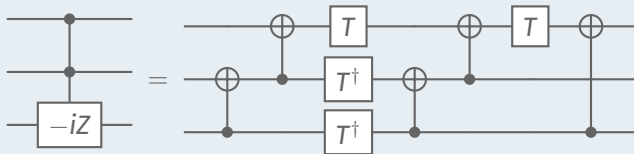
- Then apply the result from proof 1.

Many-Qubit Unitaries: Proof 3

Q: Can we implement any multi-qubit unitary using CZ and single-qubit gates? **A:** Yes!

Proof Sketch:

- First show we can implement multi-controlled Z gates with one-and-two qubit gates using (for example) the following identity:



- Then apply the Cosine-Sine decomposition inductively.

Recap: Many Physical Qubits

We have seen:

- General n -qubit states are described as vectors in $\mathbb{C}^{2^n}$. *These states are hard to describe classically.*
- Measure n -qubit states via projective measurements. *These measurements return information about the quantum state.*
- Implement any n -qubit unitary using 1- and 2-qubit gates. *These unitaries can map the $|00\dots 0\rangle$ state to an arbitrary state in $\mathbb{C}^{2^n}$.*

Question: Is this sufficient for quantum advantage? More precisely: do the arguments we have seen so far show that quantum computers can efficiently perform calculations that classical computers cannot?

Efficiently Implementable Unitaries

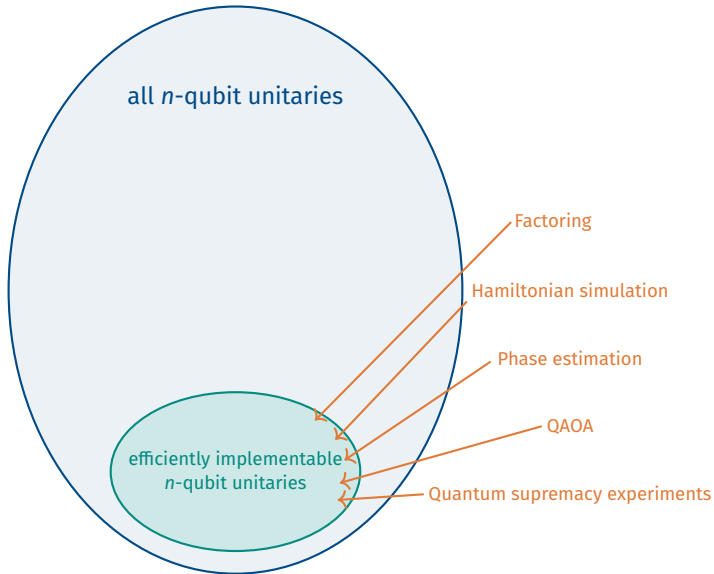
Q: Can every n -qubit unitary be implemented *efficiently* (i.e. with $\text{poly}(n)$ gates). **A:** No! In fact, most unitaries require exponentially many gates.

Proof Sketch:

- Discretize: if any unitary U_n can be implemented efficiently with $\text{poly}(n)$ one- qubit unitaries and CZ gates, then the unitary can be implemented up to error ε with single qubit unitaries described up to error $\varepsilon/\text{poly}(n)$.
- There are at least $(\frac{1}{\varepsilon})^{2^n}$ n qubit unitaries with distance ε from each other, and at most $(\text{poly}(n)/\varepsilon)^4$ different one qubit unitaries up to error $\varepsilon/\text{poly}(n)$.
- Then compute:

$$\frac{((\text{poly}(n)/\varepsilon)^4)^{\text{poly}(n)}}{(\frac{1}{\varepsilon})^{2^n}} = (\varepsilon)^{2^n} (\text{poly}(n)/\varepsilon)^{4\text{poly}(n)} \approx (\varepsilon)^{2^n}$$

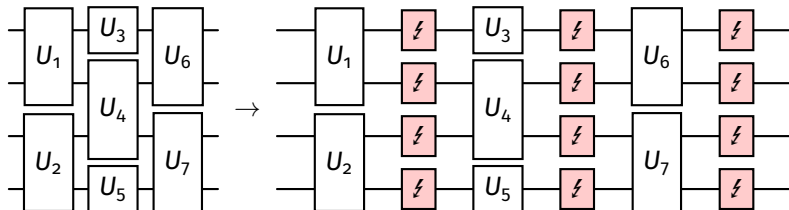
Efficiently Implementable Unitaries



Part 2: Fault-Tolerant Circuits

Motivation: Noise

Computations using **physical qubits** accumulate **noise**.



Implementing complicated unitaries (*Factoring, Phase Estimation, Hamiltonian Simulation*) requires correcting for these errors.

Fault-Tolerant Qubits

- A **fault-tolerant qubit** (or **logical qubit**) is formed from a collection of physical qubits which are continuously measured to detect errors:
 - **No error:** the measurement does not disturb the logical information stored in the qubit.
 - **Sufficiently small error:** the measurement detects the error, and a correction operation undoes the error — without disturbing the logical information.
- Because logical qubits are no longer single physical qubits evolving under a Hamiltonian, the *set of easy operations changes*:
 - **Easy:** **Pauli updates** $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$.
 - **(Usually) easy:** **Clifford gates** — unitaries that map Pauli operators to Pauli operators under conjugation:

$$\mathcal{C} = \{U : UPU^\dagger \in \mathcal{P}_n \forall P \in \mathcal{P}_n\}$$

Aside: Clifford Circuits

- Recall: a **Clifford unitary** is a unitary U satisfying $UPU^\dagger \in \mathcal{P}_n$ for all n -qubit Paulis P .
- Equivalently, Clifford unitaries are exactly the circuits generated by $\{H, CNOT, S\}$.
- Clifford circuits demonstrate many quantum effects — including entanglement and superposition. Yet they are also classically simulable!

Gottesman-Knill Theorem: Any quantum circuit consisting of:

- Preparation of computational basis states $|x\rangle, x \in \{0, 1\}^n$
 - Clifford gates
 - Intermediate Pauli measurements and classical feedforward
- can be efficiently simulated on a classical computer.

Universal Fault-Tolerant Quantum Computation

- Clifford circuits are classically simulable — to achieve quantum advantage on a fault-tolerant quantum computer we need to implement **non-Clifford gates**.
- Methods depend on the choice of error correcting code and physical qubit. One modern paradigm: implement $T = \sqrt{S}$ gates via **magic state distillation** and **state injection**.
- How many non-Clifford gates do we need?

Solovay-Kitaev Theorem: Any single-qubit unitary can be approximated to error ε using Clifford gates and $O(\log^c(1/\varepsilon))$ instances of some non-Clifford gate (or it's inverse).

Recent Improvements:

- arXiv:1403.2975 (Ross, Selinger) requires only $O(\log(1/\varepsilon))$ T gates. (Similar work by Bocharov, Roettler, Svore: arXiv:1404.5320)
- arXiv:2112.02040 (Bouland, Giurgica-Tiron) removes the need for inverses.

Part 3: Quantum Circuit Complexity

Quantum Compilation: How do we most effectively implement a target unitary?

- Target unitaries are usually useful.
- Constraints aim to model capabilities of real world (or near-term) hardware.
- Constant factors are usually important!

Quantum Circuit Complexity: What unitaries can be implemented within a given model of quantum computation?

- Computational model might not be practical!
- Constant factors often neglected.
- Interested in understanding quantum advantage and quantum-classical separations.

Sample Circuit Complexity Questions

- Can physical quantum circuits perform computation that is hard to simulate classically?
 $\rightsquigarrow$ *Quantum Supremacy*
- Can shallow (low-depth) quantum circuits perform computation that shallow classical circuits cannot?
 $\rightsquigarrow$ *Unconditional Circuit Separations*
- How hard is it to prepare a target state or implement a target unitary if classical computation and quantum arithmetic are free?
 $\rightsquigarrow$ *State Synthesis, Unitary Synthesis*

References and Further Reading

- **1- and 2-qubit unitaries, measurements**

M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*, Chapter 4. Cambridge University Press, 2000.

- **Circuit form of Cosine-Sine decomposition**

A. B. W., C. R. Chen, J. W. Helton, J. Slote, *Quantum precomputation: parallelizing cascade circuits and the Moore-Nilsson conjecture is false*, arXiv:2510.04411 (2025).

- **Quantum supremacy**

A. Harrow, A. Montanaro, *Quantum Computational Supremacy*, Nature 549, 203–209 (2017). arXiv:1809.07442.

- **Quantum advantage with shallow circuits**

S. Bravyi, D. Gosset, R. König, *Quantum advantage with shallow circuits*, Science 362, 308–311 (2018).

- **Unitary synthesis**

H. Yuen, *Lecture Notes on Unitary Synthesis*, Spring 2022.

<https://www.henryyuen.net/classes/spring2022/lec6-unitarysynthesis.pdf>